BCS-078	Simulation and Modeling	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	Define the basics of simulation modeling and replicating the practical situations in organizations
CO2	Generate random numbers and random variates using different techniques.
CO3	Develop simulation model using heuristic methods.
CO4	Analysis of Simulation models using input analyzer, and output analyzer
CO5	To learn about oriented simulation packages

Detailed Syllabus

MODULE-I

System definition and components, stochastic activities, continuous and discrete systems, system modeling, types of models, static and dynamic physical models, static and dynamic mathematical models, full corporate model, types of system study.

System simulation, why & when to simulate, nature and techniques of simulation, comparison of simulation and analytical methods, types of system simulation, real time simulation, hybrid simulation, simulation of pure-pursuit problem, single-server queuing system and an inventory problem, Monte-Carlo simulation, Distributed Lag models, Cobweb model.

MODULE-II

Simulation of continuous systems, analog vs. digital Simulation, Simulation of water reservoir system, Simulation of a servo system, simulation of an autopilot, Discrete system simulation, fixed time-step vs. even-to-even model, generation of random numbers, test for randomness, Monte-Carlo computation vs. stochastic simulation.

System dynamics, exponential growth models, exponential decay models, modified exponential growth models, logistic curves, generalization of growth models, system dynamic diagrams Introduction to SIMSCRIPT: Program, system concepts, origination, and statements, defining the telephone system model.

MODULE-III

Simulation of PERT Networks, critical path computation, uncertainties in activity duration, resource allocation and consideration. Simulation languages and software, continuous and discrete simulation languages, expression based languages, object oriented simulation, general purpose vs. application - oriented simulation packages, CSMP-III, MODSIM-III.**Course Outcomes:** After the completion of the course the student will be able to:

CO1	Describe the role of important elements of discrete event simulation and modeling paradigm.
CO2	Conceptualize real world situations related to systems development decisions, originating from source requirements and goals.
CO3	Develop skills to apply simulation software to construct and execute goal-driven system models.
CO4	Interpret the model and apply the results to resolve critical issues in a real world environment.
CO5	Understand oriented simulation packages

. **Text Books:**

1. Geoftrey Gordon, " System Simulation", PHI
2. Jerry Banks, John S. C Barry L. Nelson David M. Nicol, "Discrete Event SystemSimulation", Pearson Education

References:

1. V P Singh, "System Modeling and simulation", New Age International.
2. Averill M. Law, W. David Kelton, "System Modeling and simulation and Analysis",TMH